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#include <stdio.h>
#include <string.h>

int main()
{
    char str[20] = "helloWorld";
    char *ptr = str;
    int len = strlen(str);

    // Step 1: Modify string using pointer
    for (int i = 0; i < len; i++)
    {
        if (*(ptr + i) == 'o')
        {
            *(ptr + i) = 'O';
        }
    }
    printf("Step 1 Output: %s\n", str);
    printf("Pointer Value 1: %c\n", *ptr);
    printf("Pointer Value 2: %c\n", *(ptr + 4));
    // Step 2: Reverse the string using pointer and array
    for (int i = 0; i < len / 2; i++)
    {
        char temp = *(ptr + i);
        *(ptr + i) = *(ptr + len - i - 1);
        *(ptr + len - i - 1) = temp;
    }
    // Step 3: Print intermediate values and final string

    printf("Reversed String: %s\n", str);
    printf("Pointer Value 3: %c\n", *(ptr + len - 1));
    // Step 4: Concatenate and print
    strcat(str, "123");
    printf("After Concatenation: %s\n", str);
    printf("Pointer Value 4: %c\n", *(str + 5));
    printf("String Length: %ld\n", strlen(str));

    // Step 5: Perform pointer arithmetic
    printf("Pointer Arithmetic Result: %c\n", *(ptr + 8));
    printf("Final Character: %c\n", *(ptr+strlen(str)-1));

    return 0;
}
```